Background Bias in Image Classification:
Does a Breed Classifier Look at the Animal or at the Background?

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Abstract

Neural networks often solve a task through a *shortcut*: a feature that is correlated with the label in the training data but is not the thing we want the model to use. The background of a photo is the classic example. We study it in a small, fully controlled setting: fine-grained recognition of 37 cat and dog breeds (Oxford-IIIT Pet), where pixel-accurate pet masks let us replace the background of every test image with an unrelated scene from Places365. We define four test conditions (original, background swapped, foreground only, background only), compare a classical HOG + linear SVM baseline with five deep architectures (ResNet-18, ResNet-18 trained from scratch, ResNet-50, ConvNeXt-T, Swin-T), and test a simple remedy: replacing the background of a training image by a random scene with probability p. A ResNet-18 fine-tuned on original images loses 2.8 ± 0.7 p.p. of test accuracy when only the background changes (89.7% $\rightarrow$ 86.9%) and changes its prediction on 8.4% of the images. With background swapping ($p = 0.5$) the drop shrinks to 0.5 ± 0.2 p.p. (paired bootstrap difference 2.3 p.p., 95% CI [1.6, 2.9]) at no cost in accuracy on original images. The effect holds for all five architectures, an ablation shows that $p = 0.25$ already removes most of the gap, and among six training strategies (gray background, blurred background, CutMix, an input-gradient saliency penalty with $\lambda \in \{1, 10\}$, random scene swap) only the random scene swap removes it. Linear probes show that swap training removes most background-scene information from the learned features. The background-only condition stays close to chance for ResNet-18, so the model is *distracted* by unfamiliar backgrounds rather than reading the breed off the background. Finally, the same remedy transfers to two settings it was not designed on: real photographs from Stanford Dogs, and the Waterbirds shortcut benchmark, where it raises worst-group accuracy from 65% to 80%.

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1 Introduction

The big problem: shortcut learning. A classifier is trained to minimise its error on the training data. It will use *any* feature that lowers that error, whether or not a human would call the feature relevant. If, in the training set, huskies are mostly photographed on snow, "white background" predicts "husky" and the model may learn exactly that [15]. Geirhos et al. call this *shortcut learning* [7]: the model reaches high accuracy on data that looks like its training set and fails when the correlation breaks. The same failure appears in high-stakes applications:

- **Medicine.** Pneumonia detectors trained on chest X-rays from several hospitals learned to recognise the *hospital* (from scanner-specific tokens and image statistics), which correlated with disease prevalence, and generalised worse to new hospitals [22].
- **Ecology.** Animal detectors for camera traps fail at new camera locations because they rely on the scenery of the locations they were trained on [2].
- **Agriculture and remote sensing.** Plant-disease and land-use classifiers face the same risk when the field, the lighting or the region is correlated with the class.

A shortcut is invisible on a test set drawn from the same distribution as the training set: the correlation is there as well, so the shortcut "works". To see it, one has to *break the correlation on purpose* and measure what happens. That is what this project does.

Our small, controllable case. We pick a task where the shortcut can be switched on and off precisely: breed classification on Oxford-IIIT Pet [14]. Every image comes with a *trimap* that marks the pet, the background and an uncertain border. With the mask we can paste the pet onto a different background, remove the background entirely, or remove the pet and keep only the background. Breeds are a fine-grained task (37 classes, some of them almost identical), so a model has reasons to grab any extra cue, and the background of a pet photo (sofa, lawn, show ring) is a plausible one.

Research questions.

RQ1. Does a standard fine-tuned CNN rely on the background to classify the breed?

RQ2. Does randomly replacing the background during training reduce that reliance, and what does it cost in ordinary accuracy?

RQ3. Does the answer depend on the architecture, and how does the remedy compare with other ways of handling the background?

RQ4. Does a remedy learned on synthetic composites transfer to real photographs and to a standard shortcut benchmark?

Hypotheses (written down before training).

H1. ResNet-18 A (trained on original images) loses noticeable accuracy when the background is swapped.

H2. ResNet-18 B (trained with random background swap) has a smaller gap than A, at comparable accuracy on original images.

H3. Both models lose less on foreground-only images (neutral gray background) than on swapped images: a neutral background carries no misleading signal, while a foreign scene can actively mislead.

Contributions. (i) A controlled evaluation protocol with four test conditions and fixed, reproducible test backgrounds drawn from scene categories that never occur in training. (ii) A comparison of 17 trained configurations, 51 runs in total (3 seeds each), plus a classical baseline.

(iii) Statistical tests (paired bootstrap) for the main claims. (iv) Mechanistic analyses: per-breed gaps, Grad-CAM, and linear probes of background information in the features. (v) Two transfer studies: an external test on real photos (Stanford Dogs) and a case study on Waterbirds.

2 Related work

Backgrounds in object recognition. Xiao et al. [20] built ImageNet-9 with variants such as *Only-FG*, *Mixed-Same* (foreground on a background of the same class) and *Mixed-Rand* (foreground on a random background) and showed that ImageNet models depend on backgrounds, less so for more accurate models. Our "swapped" condition is their Mixed-Rand, our "foreground only" is their Only-FG (gray instead of black), and our "background only" corresponds to their Only-BG-T. Our dataset differs: a fine-grained task with exact masks.

Shortcut learning and spurious correlations. Geirhos et al. [7] give the general framing. Beery et al. [2] show the location dependence of camera-trap models. Sagawa et al. [17] introduce Waterbirds (CUB birds [19] pasted onto Places scenes with a 95% bird–background correlation) and the *worst-group accuracy* metric; they fix it with group DRO, which requires group labels. Our remedy needs foreground masks instead of group labels.

Fixing shortcuts. Data-side fixes change what the model sees: background randomisation (ours, and Mixed-Rand training in [20]), or generic augmentation such as CutMix [21]. Loss-side fixes change what the model is rewarded for: "Right for the Right Reasons" [16] penalises input gradients on regions annotated as irrelevant. We compare both families under one protocol.

Tools. ResNet [8], ConvNeXt [11], Swin Transformer [10], all pretrained on ImageNet [5]; Places365 [23]; HOG [4]; Grad-CAM [18]; linear probes [1]; DeepLabV3 [3]; AdamW [13]; cosine schedules [12]; bootstrap [6]; Stanford Dogs [9].

3 Data

3.1 Oxford-IIIT Pet

Oxford-IIIT Pet [14] has 7,349 photos of 37 breeds (12 cat, 25 dog breeds), about 200 per breed, with an official `trainval/test` split. Each image has a trimap $T(u) \in \{1, 2, 3\}$ for every pixel u: 1 = pet, 2 = background, 3 = border/uncertain (usually a band of fur along the outline).

Splits. The official `test` split (3,669 images) is our test set and was used once, at the very end. We split the official `trainval` (3,680 images) into train and validation with a stratified (by breed) 85/15 split, `random_state=42`. Splits are made on image ids before any variant exists, so all versions of an image stay in the same split; the code asserts that the three sets are disjoint and that every split contains all 37 classes. The CSV files are committed to the repository and used by every model.

Split	Images	Cats	Dogs	Min / max images per breed
Train	3,128	1,010	2,118	79 / 85
Validation	552	178	374	14 / 15
Test (official)	3,669	1,183	2,486	88 / 100

Table 1: Pet splits. Classes are balanced, so chance accuracy is $1/37 = 2.7\%$.

Data audit. All 7,349 images and trimaps load; 3 images are RGBA and are converted to RGB; every trimap matches its image size. On average the pet covers 30.0% of an image (class 1), the border 11.8% (class 3) and the background 58.2% (class 2); the pet fraction ranges from 0% to 94.7%, so a few masks are nearly empty.

3.2 Background scenes: Places365

Backgrounds come from the Places365 validation set [23] (36,500 images at 256×256, 100 per category). We first drop the 17 categories in which animals often appear (every category whose name contains one of *kennel, pasture, corral, barn, stable, veterinarian, pet_shop, zoo, aquarium, ranch, farm, chicken, cattle, hunting, fishpond, field/wild, rodeo, racecourse*), so that a pasted pet rarely lands next to another animal. The remaining 348 categories are shuffled (seed 42) and *whole categories* are assigned to three disjoint pools:

Pool	Categories	Images	Used for
`bg_train`	209	20,900	random swaps during training (models B, ablation, Waterbirds B)
`bg_val`	52	5,200	swapped validation set (monitoring only)
`bg_test`	87	8,700	swapped test set; probing labels

Table 2: Background pools. Pools share no image and no scene category, so test backgrounds are *new types of scene*, not just new photos of known scenes.

Fixed evaluation backgrounds. For every validation and test image we draw one background from its pool once (seed fixed) and store it in a manifest (`manifests/{val,test}_swapped.csv`). Composites are rendered on the fly but deterministically, so every model is evaluated on *exactly the same* swapped images.

3.3 Composition: how a swapped image is made

Let $x \in [0, 255]^{H \times W \times 3}$ be the photo and T its trimap. The binary foreground is

$$m(u) = \mathbf{1}\big[T(u) = 1\big] \;\vee\; \mathbf{1}\big[T(u) = 3\big], \tag{1}$$

i.e. the border class is counted as pet. Without it, the outer fur would be cut off and the pet would look "shaved". A hard mask makes a visible seam, so we soften it with a Gaussian blur of $\sigma = 1.5$ px:

$$\alpha = G_\sigma * m \;\in [0, 1]^{H \times W}. \tag{2}$$

A new background b is scaled to cover $H \times W$ (aspect preserved) and cropped at relative offset $o \in [0, 1]^2$ (the centre $o = (0.5, 0.5)$ at test time, uniformly random in training). The composite is standard alpha blending, pixel by pixel:

$$\tilde{x}(u) = \alpha(u)\, x(u) + \big(1 - \alpha(u)\big)\, b(u). \tag{3}$$

Pixels with $\alpha = 1$ keep the pet, pixels with $\alpha = 0$ show the new scene, and the blurred band in between mixes the two, which hides the seam.

3.4 The four test conditions

All four are built from the same test image with Eq. (3); only b (and, for the last one, the roles of the pet and the background) change. Let g be a constant image with the mean ImageNet colour $(124, 116, 104)$.

Condition	Definition	What it tests
original	x unchanged	ordinary accuracy
swapped	$\alpha x + (1 - \alpha)\, b_{\text{test}}$, b_{test} from the manifest	robustness to a foreign scene
fg-only	$\alpha x + (1 - \alpha)\, g$	accuracy without any background
bg-only	$\alpha g + (1 - \alpha)\, x$	can the breed be read off the background alone?

Why fg-only matters. A drop on swapped images could have two causes: (a) the model relied on the original background, or (b) the model is hurt by *paste artifacts* (seams, lighting mismatch). Fg-only images have the same paste artifacts but no misleading scene. If the fg-only drop is much smaller than the swapped drop, artifacts are not the main cause (H3). **Why bg-only matters.** If a model scores far above chance with the pet painted over, the background (or the pet's silhouette, which bg-only keeps) carries breed information that the model uses.

Figure 10 in the appendix shows examples of all conditions.

3.5 External data

Stanford Dogs [9] has 20,580 photos of 120 breeds with bounding boxes. Twenty-one of its breeds are also Pet breeds (mapping by ImageNet synset in `src/bgbias/stanford_dogs.py`; the pit bull is left out because Stanford has only the American Staffordshire terrier, and American bulldog, Havanese and Shiba Inu have no counterpart). This gives 3,730 real photos, taken by other people in other places, with no cut-and-paste. We evaluate two views: *full* (whole photo) and *crop* (the union of all annotated dog boxes, which removes most of the background).

Waterbirds [17]: CUB-200 birds [19] pasted onto Places scenes, label $y \in \{\text{landbird}, \text{waterbird}\}$, place $a \in \{\text{land}, \text{water}\}$. In train, 95% of waterbirds sit on water and 95% of landbirds on land; the test set has all four (bird, place) groups. Splits: 4,795 train, 1,199 val, 5,794 test. Waterbirds has no masks, so we segment the bird with a pretrained DeepLabV3-ResNet50 [3] (class "bird" of the VOC label set) and use that mask in Eq. (3).

4 Methods

4.1 Notation and the learning problem

A model f_θ maps an image to logits $z = f_\theta(x) \in \mathbb{R}^K$, $K = 37$. Class probabilities are the softmax

$$p_\theta(k \mid x) = \frac{\exp z_k}{\sum_{j=1}^{K} \exp z_j}, \qquad \hat{y}(x) = \arg\max_k z_k. \tag{4}$$

All networks are trained with the cross-entropy loss on a mini-batch $\mathcal{B}$:

$$\mathcal{L}_{\mathrm{CE}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{(x,y)\in\mathcal{B}} \log p_\theta(y \mid x). \tag{5}$$

4.2 The remedy: random background swap (model B)

Each time a training image is loaded, with probability p its background is replaced by a random scene:

$$x' = \begin{cases} \alpha\, x + (1-\alpha)\, b, & \text{with probability } p, \quad b \sim \text{Uniform}(\texttt{bg_train}),\ o \sim U[0,1]^2, \\ x, & \text{with probability } 1-p, \end{cases} \tag{6}$$

and then the usual augmentations are applied to x' (random resized crop, horizontal flip). A new b is drawn every epoch, so the model sees each pet on many scenes. The intuition: if the background is random, it is independent of the label, $b \perp y$, so it cannot lower the training loss, and gradient descent has no reason to use it. Keeping $1-p$ of the images original keeps the training distribution close to the test distribution of real photos. **Model A** is the same model, same code, same hyperparameters, with $p = 0$. Within an architecture, the configuration files of A and B differ only in `name` and `p_swap`.

4.3 Alternative ways to handle the background (all on ResNet-18)

- **Gray background** ($p = 0.5$): $b = g$ in Eq. (6). Removes the background instead of randomising it.
- **Blurred background** ($p = 0.5$): $b = G_{12} * x$, the image's own background blurred with radius 12 px. Keeps the colours of the original scene but removes its detail.
- **CutMix** [21]: a generic augmentation that does *not* use masks. With probability 0.5 per batch, each image x_i is paired with another image x_j of the same batch (the batch shifted by one) and a random box R is chosen; the box of x_i is replaced by the same box of x_j and the labels are mixed by area:

$$\tilde{x}_i = (1 - M_R) \odot x_i + M_R \odot x_j, \qquad \tilde{y}_i = \lambda\, e_{y_i} + (1 - \lambda)\, e_{y_j}, \qquad \lambda = 1 - \frac{|R|}{HW}, \tag{7}$$

  with $\lambda \sim \mathrm{Beta}(1, 1)$ before box placement (`torchvision.transforms.v2.CutMix` defaults). The loss is the cross-entropy with the soft target $\tilde{y}_i$. CutMix tests whether a generic regulariser is enough.
- **Saliency penalty** ("Right for the Right Reasons", RRR [16]): train on original images only, but penalise sensitivity to background pixels. Let

$$G = \nabla_x \sum_{k=1}^{K} \log p_\theta(k \mid x) \in \mathbb{R}^{3\times H\times W}, \qquad E(u) = \sum_{c=1}^{3} G_c(u)^2 \tag{8}$$

  be the input gradient and its energy per pixel. The penalty is the share of that energy on the background,

$$S(x) = \frac{\sum_u E(u)\,(1 - \alpha(u))}{\sum_u E(u)} \in [0, 1], \qquad \mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda\, \frac{1}{|\mathcal{B}|} \sum_{x\in\mathcal{B}} S(x), \tag{9}$$

  with $\lambda \in \{1, 10\}$. Because S is a fraction, λ is on the same scale as the cross-entropy. (Ross et al. penalise $\sum_u (1 - \alpha)E$ directly; we normalise so that the size of the gradient does not matter, only where it points.) Minimising S needs the derivative of a gradient ("double backpropagation"), which is about twice as expensive; these runs use fp32, because the input gradients underflow in fp16. The image and its mask go through the *same* random crop and flip (`torchvision.transforms.v2` joint transform).

4.4 Architectures

Model	Type	Params	Initialisation (torchvision)
ResNet-18 [8]	CNN, residual	11.7 M	`IMAGENET1K_V1`
ResNet-18 scratch	same	11.7 M	random (Kaiming)
ResNet-50 [8]	CNN, bottleneck residual	25.6 M	`IMAGENET1K_V2`
ConvNeXt-T [11]	modern CNN (large kernels, LayerNorm)	28.6 M	`IMAGENET1K_V1`
Swin-T [10]	vision transformer, shifted windows	28.3 M	`IMAGENET1K_V1`

In every model the ImageNet classifier is replaced by a new linear layer with 37 outputs; the rest of the network (the *backbone*) is fine-tuned. The scratch model shows what happens without ImageNet knowledge: it has to learn everything from 3,128 images.

4.5 Training protocol

AdamW. With gradient g_t, AdamW keeps running averages of the gradient and its square,

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad \theta_t = \theta_{t-1} - \eta_t \Big(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + w\, \theta_{t-1} \Big), \tag{10}$$

where $\hat{m}_t, \hat{v}_t$ are the bias-corrected averages and w the weight decay, applied directly to the weights ("decoupled"), not through the gradient as in Adam with L2 [13].

Setting	Value
Input	224×224, ImageNet mean/std normalisation
Train augmentation	(background swap, Eq. 6) $\rightarrow$ RandomResizedCrop(224, scale $[0.7, 1]$) $\rightarrow$ horizontal flip
Eval transform	resize shorter side to 256 $\rightarrow$ centre crop 224
Optimizer	AdamW, $\beta = (0.9, 0.999)$, weight decay 10^{-4}
Learning rate	backbone 10^{-4}, new head 10^{-3} (two parameter groups)
Schedule	cosine annealing over all steps, no warm-up, $\eta_{\min} = 0$
Epochs / batch	15 epochs, batch 64 ($\approx$49 steps per epoch, 735 steps)
Precision	mixed precision (fp16 autocast + gradient scaling); fp32 for RRR
Checkpoint	epoch with the best *original* validation accuracy (same rule for A and B)
Seeds	0, 1, 2 (Python, NumPy, PyTorch, CUDA; DataLoader generator reseeded every epoch)
Scratch ResNet-18	40 epochs, one learning rate 10^{-3} for all layers, weight decay $5 \cdot 10^{-2}$
Waterbirds	ResNet-18, 2 classes, 10 epochs, otherwise as above; checkpoint by validation *average* accuracy

Table 3: Training settings. They were fixed on the validation set for ResNet-18 A and then reused unchanged for every configuration; nothing was tuned on the test set.

Cosine schedule. At step t of $T_{\max}$ (all steps of all epochs),

$$\eta_t = \eta_{\min} + \tfrac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\pi t / T_{\max})), \tag{11}$$

so the learning rate starts at η_0 and decays smoothly to 0; each parameter group has its own η_0.

Why two learning rates. The backbone already holds useful ImageNet features; a large step would destroy them. The new head starts from random weights and needs to move fast. A $10\times$ larger rate for the head is the standard fine-tuning compromise.

Checkpoint selection and the validation sets. After every epoch we compute validation accuracy on original images and on swapped images (backgrounds from `bg_val`). The swapped validation accuracy is *only logged*; the checkpoint is always chosen by original validation accuracy, for A and B alike. Choosing by swapped accuracy would select for robustness and would favour B unfairly.

Reproducibility. All randomness (background choice, crop offset, augmentations, shuffling) comes from PyTorch generators seeded from (seed, epoch), so an interrupted run resumed from its checkpoint continues exactly as an uninterrupted one. The baseline A/B runs were first trained in Google Colab (T4 GPU) and later retrained from scratch on a local GPU with the same code (test metrics agree within 0.3 p.p.; e.g. gap of A: 2.8 $\rightarrow$ 2.8 p.p., gap of B: 0.6 $\rightarrow$ 0.5 p.p.).

4.6 Classical baseline: HOG + linear SVM

Images are resized to 128×128 gray, and HOG [4] features are computed with 9 orientation bins, 8×8-pixel cells and 2×2-cell blocks, giving $15 \cdot 15 \cdot 2 \cdot 2 \cdot 9 = 8{,}100$ features. With 3,128 training images a linear SVM on 8,100 raw features separates the training set perfectly and overfits, so the pipeline is StandardScaler $\rightarrow$ PCA (256 components) $\rightarrow$ LinearSVC. C was chosen on validation accuracy from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$; the best was $C = 10^{-5}$ (validation accuracy 9.1%), at the edge of the grid. The baseline is trained once on original images and evaluated on the same four conditions. It is a lower bound that shows how hard the task is without learned features.

5 Evaluation metrics and analysis methods

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be the test set ($N = 3{,}669$) and $c \in \{\text{orig}, \text{swap}, \text{fg}, \text{bg}\}$ a condition; $x_i^{(c)}$ is image i under condition c.

Accuracy and macro F1.

$$\text{Acc}_c = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}(x_i^{(c)}) = y_i], \qquad \text{F1}_c^{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} \frac{2P_k R_k}{P_k + R_k}, \tag{12}$$

where P_k and R_k are precision and recall of class k. Classes are balanced, so the two agree closely (Table 13); we report accuracy in the main text.

Robustness gap. The accuracy lost when only the background changes:

$$\text{Gap}_{\text{swap}} = \text{Acc}_{\text{orig}} - \text{Acc}_{\text{swap}}, \qquad \text{Gap}_{\text{fg}} = \text{Acc}_{\text{orig}} - \text{Acc}_{\text{fg}}. \tag{13}$$

It is reported in percentage points (p.p.). A gap must be read together with the accuracy: a weak model can have a small gap simply because it is wrong everywhere.

Flip rate. The share of images whose prediction changes when only the background changes:

$$\text{Flip}_{\text{swap}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}(x_i^{(\text{swap})}) \neq \hat{y}(x_i^{(\text{orig})})]. \tag{14}$$

The gap counts only net changes: an image that becomes wrong and another that becomes right cancel out. The flip rate counts both, so Flip $\geq |\text{Gap}|$ and it measures how *stable* the decision is, regardless of correctness.

Mean and spread over seeds. Every deep configuration is trained with 3 seeds s; tables show $\bar{m} \pm s_m$ with the sample standard deviation $s_m = \sqrt{\frac{1}{n-1}\sum_s (m_s - \bar{m})^2}$, $n = 3$. This spread measures training randomness (initialisation of the head, data order, augmentations), not test-set sampling noise.

Paired bootstrap for the difference of gaps. To test H2 we need the uncertainty of $\text{Gap}(A) - \text{Gap}(B)$ due to the finite test set. For every test image i, average over seeds the per-image gap

$$d_i^M = \frac{1}{3}\sum_s \Big(\mathbf{1}[\hat{y}^{M,s}(x_i^{(\text{orig})}) = y_i] - \mathbf{1}[\hat{y}^{M,s}(x_i^{(\text{swap})}) = y_i]\Big), \qquad \Delta_i = d_i^A - d_i^B, \tag{15}$$

so that $\bar{\Delta} = \frac{1}{N}\sum_i \Delta_i = \text{Gap}(A) - \text{Gap}(B)$. We draw $n_{\text{boot}} = 2{,}000$ bootstrap samples of N indices with replacement, compute the mean $\bar{\Delta}^*$ of each resample, and report the 2.5th and 97.5th percentiles as a 95% confidence interval [6]. The bootstrap is *paired*: A and B are resampled on the same images, so image difficulty cancels out. The same procedure gives confidence intervals for accuracy differences on Stanford Dogs.

Grad-CAM and the share of attention on the pet. For ResNet-18, Grad-CAM [18] uses the activations $A^k \in \mathbb{R}^{7\times 7}$ of the k-th channel of the last convolutional block (`layer4`) and the gradient of the predicted class score $z_{\hat{y}}$:

$$w_k = \frac{1}{49}\sum_{i,j} \frac{\partial z_{\hat{y}}}{\partial A_{ij}^k}, \qquad \text{CAM} = \text{ReLU}\Big(\sum_k w_k A^k\Big), \tag{16}$$

upsampled bilinearly to 224×224 and scaled to $[0, 1]$. We summarise a map by the share of its mass on the pet,

$$\text{CAM-on-pet} = \frac{\sum_u \text{CAM}(u)\,\alpha(u)}{\sum_u \text{CAM}(u)}, \tag{17}$$

using the mask cropped exactly like the image. It is computed on 500 test images for every seed; the pet covers on average 0.59 of those crops, so a map that ignored the mask would score about 0.59.

Linear probe of background information. How much does a model's internal representation know about the background? We take the penultimate features $h(x)$ (the input of the classifier) of every *swapped* test image and train a linear classifier to predict the Places365 *scene category* of its background (87 categories, chance $1/87 = 1.1\%$). The probe is StandardScaler $\rightarrow$ PCA(128) $\rightarrow$ multinomial logistic regression ($C = 0.5$), scored by 5-fold stratified cross-validation. Test backgrounds were assigned at random, so they carry no information about the breed: a model that keeps this information in its features spends capacity on something useless for the task, which is the representational side of background reliance [1]. As references we probe the same architectures with ImageNet weights before fine-tuning and a randomly initialised ResNet-18.

External test on Stanford Dogs. Each model predicts over all 37 Pet classes (a cat prediction is an error). We also report a dog-restricted accuracy where the argmax is taken over the 21 present breeds only. *Caveat:* Stanford Dogs is built from ImageNet images, so ImageNet-pretrained backbones have seen these photos (with ImageNet labels) during pretraining. This inflates absolute numbers but affects A and B equally, so the A–B comparison is fair; the scratch models never saw them.

Worst-group accuracy (Waterbirds). With groups $g = (y, a)$,

$$\text{Acc}_g = \frac{1}{|\mathcal{D}_g|} \sum_{i \in \mathcal{D}_g} \mathbf{1}[\hat{y}_i = y_i], \qquad \text{WGA} = \min_g \text{Acc}_g, \qquad \text{Avg} = \frac{1}{N} \sum_i \mathbf{1}[\hat{y}_i = y_i]. \tag{18}$$

A model that uses the place as a shortcut fails on the rare groups (waterbird on land, landbird on water), so WGA is the number that exposes it. Group labels are used only for evaluation, never for training or checkpoint selection.

6 Results

6.1 Main experiment: ResNet-18 A vs B (H1–H3)

H1 confirmed. ResNet-18 A drops from 89.7% to 86.9% on swapped images, a gap of 2.8 ± 0.7 p.p., and 8.4% of its predictions change (Table 4, Fig. 1). The pet is pixel-for-pixel identical in both images; only the scene behind it differs.

H2 confirmed. ResNet-18 B has the same original accuracy (89.7%) but a gap of only 0.5 ± 0.2 p.p. and a flip rate of 3.7%, less than half of A's. The paired bootstrap gives $\text{Gap}(A) - \text{Gap}(B) = 2.3$ p.p., 95% CI [1.6, 2.9]; none of the 2,000 bootstrap samples is ≤ 0. Background swapping costs nothing on original images.

H3 confirmed. Both models lose less on fg-only images than on swapped ones (A: 1.1 vs 2.8 p.p.; B: 0.2 vs 0.5 p.p.). Fg-only images have the same paste seams as swapped ones, so A's swapped drop is mostly caused by the foreign scene, not by artifacts.

Model	Seeds	Test accuracy (%)				Gap (p.p.)		Flip
		Original	Swapped	Fg-only	Bg-only	swap	fg	swap (%)
HOG + linear SVM	1	9.8 ± 0.0	8.5 ± 0.0	7.3 ± 0.0	3.8 ± 0.0	1.3 ± 0.0	2.5 ± 0.0	74.3 ± 0.0
ResNet-18 A ($p = 0$)	3	89.7 ± 0.3	86.9 ± 0.6	88.6 ± 0.8	3.6 ± 0.5	2.8 ± 0.7	1.1 ± 0.8	8.4 ± 0.9
ResNet-18, $p = 0.25$	3	89.5 ± 0.6	89.0 ± 0.8	89.1 ± 0.8	3.4 ± 0.2	0.4 ± 0.3	0.4 ± 0.4	4.4 ± 0.4
ResNet-18 B ($p = 0.5$)	3	89.7 ± 0.2	89.1 ± 0.0	89.4 ± 0.4	3.0 ± 0.2	0.5 ± 0.2	0.2 ± 0.3	3.7 ± 0.2
ResNet-18, $p = 1.0$	3	88.8 ± 0.3	88.9 ± 0.5	88.8 ± 0.3	3.2 ± 0.0	−0.1 ± 0.2	−0.0 ± 0.3	4.4 ± 0.5
ResNet-18, gray background	3	89.5 ± 0.4	88.1 ± 1.0	89.6 ± 0.6	3.5 ± 0.2	1.4 ± 0.7	−0.1 ± 0.2	7.3 ± 1.1
ResNet-18, blurred background	3	89.2 ± 1.0	87.3 ± 1.0	88.7 ± 1.1	3.6 ± 0.5	2.0 ± 0.5	0.5 ± 0.2	8.3 ± 0.5
ResNet-18, CutMix	3	87.7 ± 0.5	82.7 ± 0.9	85.4 ± 0.7	3.5 ± 0.2	5.0 ± 0.8	2.3 ± 0.2	14.3 ± 0.8
ResNet-18, saliency penalty $\lambda = 1$	3	88.7 ± 0.4	86.4 ± 0.3	87.8 ± 0.2	4.1 ± 0.3	2.3 ± 0.5	0.9 ± 0.2	7.7 ± 0.7
ResNet-18, saliency penalty $\lambda = 10$	3	84.5 ± 1.6	81.5 ± 2.4	83.1 ± 2.2	3.6 ± 0.2	3.0 ± 0.8	1.4 ± 0.6	11.9 ± 2.1
ResNet-18 scratch A	3	50.3 ± 1.6	42.4 ± 1.4	49.6 ± 1.1	3.9 ± 0.3	7.9 ± 0.2	0.7 ± 0.5	45.0 ± 1.1
ResNet-18 scratch B	3	58.7 ± 0.9	58.0 ± 1.2	60.5 ± 1.5	3.6 ± 1.0	0.7 ± 0.3	−1.8 ± 0.8	21.6 ± 0.9
ResNet-50 A	3	91.7 ± 0.3	90.9 ± 0.2	91.3 ± 0.5	4.8 ± 0.3	0.8 ± 0.2	0.4 ± 0.2	4.5 ± 0.1
ResNet-50 B	3	92.0 ± 0.5	91.8 ± 0.3	91.8 ± 0.3	4.4 ± 0.6	0.3 ± 0.2	0.2 ± 0.2	2.1 ± 0.2
ConvNeXt-T A	3	94.4 ± 0.4	93.4 ± 0.5	94.0 ± 0.4	9.6 ± 0.8	1.0 ± 0.1	0.4 ± 0.2	3.1 ± 0.5
ConvNeXt-T B	3	94.3 ± 0.2	94.0 ± 0.3	94.2 ± 0.2	9.2 ± 0.3	0.3 ± 0.1	0.1 ± 0.0	1.6 ± 0.2
Swin-T A	3	93.5 ± 0.2	91.9 ± 0.4	93.2 ± 0.2	7.4 ± 0.9	1.6 ± 0.3	0.4 ± 0.2	5.5 ± 0.4
Swin-T B	3	93.8 ± 0.2	93.5 ± 0.4	93.5 ± 0.3	5.7 ± 0.9	0.3 ± 0.2	0.3 ± 0.3	2.1 ± 0.1

Table 4: All configurations on the Pet test set (mean ± std over 3 seeds; HOG is deterministic). Chance accuracy is 2.7%. Gap = original − condition; flip = share of predictions that change.

Background only is near chance. With the pet painted over, A reaches 3.6% and B 3.0%, barely above chance (2.7%). So the ResNet-18 does *not* read the breed off the background. Together with H1 this means the model is *distracted* by an unfamiliar background (the scene pushes the features away from where the classifier expects them) rather than exploiting a learned breed–background association. Note that this is the more benign kind of background reliance; in Waterbirds (Section 6.9) the background *is* predictive and the failure is much larger.

HOG + SVM reaches only 9.8% on original images (3.6× chance), confirming that the task needs learned features. Its flip rate of 74% shows that its predictions are essentially unstable; its small gap (1.3 p.p.) is meaningless at this accuracy.

6.2 How much swapping is needed? Ablation over p

p	Original (%)	Swapped (%)	Fg-only (%)	Gap swap (p.p.)	Flip (%)
0.0	89.7 ± 0.3	86.9 ± 0.6	88.6 ± 0.8	2.8 ± 0.7	8.4 ± 0.9
0.25	89.5 ± 0.6	89.0 ± 0.8	89.1 ± 0.8	0.4 ± 0.3	4.4 ± 0.4
0.5	89.7 ± 0.2	89.1 ± 0.0	89.4 ± 0.4	0.5 ± 0.2	3.7 ± 0.2
1.0	88.8 ± 0.3	88.9 ± 0.5	88.8 ± 0.3	−0.1 ± 0.2	4.4 ± 0.5

Table 5: ResNet-18 with swap probability p (3 seeds each); $p = 0$ is A, $p = 0.5$ is B.

Already $p = 0.25$ removes almost the whole gap (2.8 → 0.4 p.p., Table 5, Fig. 2). Original accuracy stays within ±0.6 p.p., which is the size of the seed spread, up to $p = 0.5$; at $p = 1$ (the model never sees a real background) it is 0.9 p.p. lower and the gap is slightly negative (−0.1 p.p.): the model is now *better* on composites than on real photos, because composites are its training distribution. $p = 0.5$ is the best compromise: highest original accuracy and lowest flip rate.

6.3 Does it depend on the architecture?

Background swapping lowers the gap for every architecture, and every bootstrap interval excludes zero (Table 6, Fig. 3). Three patterns stand out:

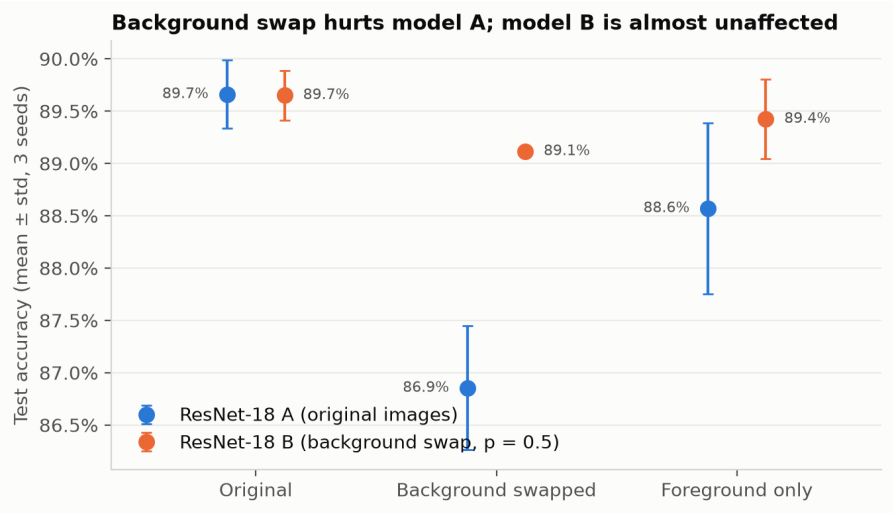


Figure 1: Test accuracy of ResNet-18 A and B under three conditions (mean ± std over 3 seeds). Both models are equally accurate on original images; only A loses accuracy when the background is swapped.

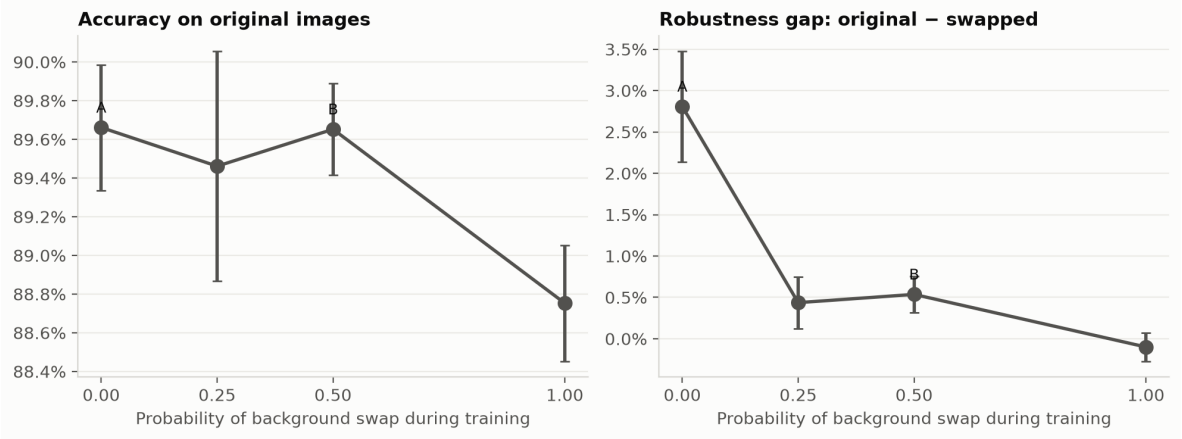


Figure 2: Accuracy and gap as a function of the swap probability p.

1. **Stronger models depend less on the background.** The gap of A falls from 2.8 p.p. (ResNet-18) to 0.8–1.6 p.p. for ResNet-50, ConvNeXt-T and Swin-T, matching the observation of Xiao et al. that more accurate models are less background-sensitive [20]. B still cuts the remaining gap by a factor of 3–5 and halves the flip rate.
2. **Without pretraining the dependence is largest.** The scratch ResNet-18 is a weak model (50%) with a 7.9 p.p. gap and a 45% flip rate: without ImageNet features, the background is an easy statistic to fit. Here swapping is also a strong regulariser: B improves original accuracy by 8.4 p.p. and reduces the gap to 0.7 p.p. Its fg-only accuracy is even higher than its original accuracy.
3. **Bg-only rises with model strength.** ConvNeXt-T and Swin-T reach 7–10% on bg-only images, 3–4× chance, and swapping barely lowers it. Since bg-only images keep the *silhouette* of the pet (the painted region has its exact shape), and swapping does not remove silhouette information, we attribute most of this to shape, not scenery (see Section 7).

6.4 Comparison of strategies for handling the background

Only the random scene swap removes the gap (Table 7, Fig. 4):

- **Gray** (1.4 p.p.) and **blurred** (2.0 p.p.) backgrounds help partially. They teach the model to work without a background, but never show it a *misleading* one, which is what the swapped

Architecture		Original (%)	Swapped (%)	Gap (p.p.)	Flip (%)	Bg-only (%)	Gap A−B [95% CI]
ResNet-18	A	89.7 ± 0.3	86.9 ± 0.6	2.8 ± 0.7	8.4 ± 0.9	3.6	
	B	89.7 ± 0.2	89.1 ± 0.0	0.5 ± 0.2	3.7 ± 0.2	3.0	2.3 [1.6, 2.9]
ResNet-18 (scratch)	A	50.3 ± 1.6	42.4 ± 1.4	7.9 ± 0.2	45.0 ± 1.1	3.9	
	B	58.7 ± 0.9	58.0 ± 1.2	0.7 ± 0.3	21.6 ± 0.9	3.6	7.2 [6.0, 8.5]
ResNet-50	A	91.7 ± 0.3	90.9 ± 0.2	0.8 ± 0.2	4.5 ± 0.1	4.8	
	B	92.0 ± 0.5	91.8 ± 0.3	0.3 ± 0.2	2.1 ± 0.2	4.4	0.5 [0.1, 1.0]
ConvNeXt-T	A	94.4 ± 0.4	93.4 ± 0.5	1.0 ± 0.1	3.1 ± 0.5	9.6	
	B	94.3 ± 0.2	94.0 ± 0.3	0.3 ± 0.1	1.6 ± 0.2	9.2	0.7 [0.3, 1.1]
Swin-T	A	93.5 ± 0.2	91.9 ± 0.4	1.6 ± 0.3	5.5 ± 0.4	7.4	
	B	93.8 ± 0.2	93.5 ± 0.4	0.3 ± 0.2	2.1 ± 0.1	5.7	1.3 [0.8, 1.9]

Table 6: Five architectures, A ($p = 0$) vs B ($p = 0.5$), 3 seeds each. The last column is the paired-bootstrap difference of gaps.

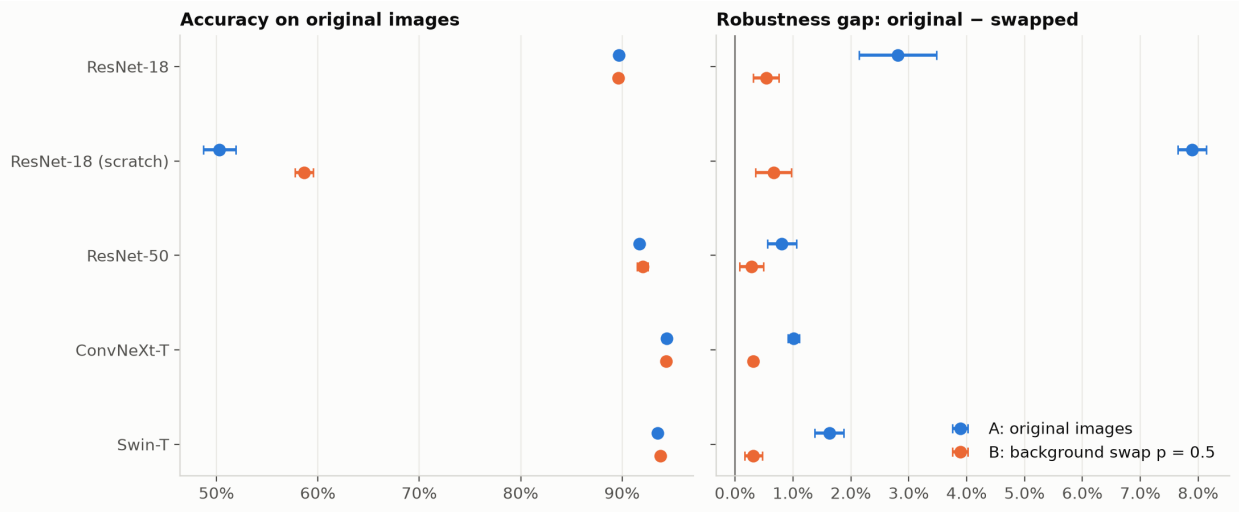


Figure 3: Original accuracy (left) and robustness gap (right) for five architectures.

test contains.

- **CutMix makes it worse** (gap 5.0 p.p., flip 14.3%, original −2.0 p.p.). CutMix pastes rectangles from other images *over* the pet too and mixes the labels by area, so it does not separate pet from background; it seems to make the model more sensitive to local content anywhere in the image.
- **The saliency penalty does not help**: $\lambda = 1$ gives 2.3 p.p. (within the seed spread of A), $\lambda = 10$ gives 3.0 p.p. and costs 5.2 p.p. of original accuracy. The penalty is minimised during training (for seed 0 the background share of the gradient energy is 6.9% for $\lambda = 1$ and 3.5% for $\lambda = 10$ in the last epoch, against a background area of about 60%), but a small input gradient on the background is a *local* property at the training images; it does not stop a large, structured change of the background from moving the features.

6.5 Which breeds depend on the background?

The largest effect is a *pair*: Staffordshire bull terrier (+28 p.p.) and American pit bull terrier (−28 p.p.). These two breeds look almost the same, and on swapped images A moves many Staffordshire terriers to "pit bull". So for visually ambiguous breeds the background *was* tipping the decision. B shrinks both to +0.7 and −5.7 p.p. On average cats have a larger gap than dogs for A (4.1 vs 2.3 p.p.; B: 1.1 vs 0.2 p.p.), e.g. Sphynx (+11.3 p.p.) and Abyssinian (+7.5 p.p.).

Method (ResNet-18)	Original (%)	Swapped (%)	Fg-only (%)	Gap (p.p.)	Flip (%)
None (A)	89.7 ± 0.3	86.9 ± 0.6	88.6 ± 0.8	2.8 ± 0.7	8.4 ± 0.9
CutMix	87.7 ± 0.5	82.7 ± 0.9	85.4 ± 0.7	5.0 ± 0.8	14.3 ± 0.8
Blurred background	89.2 ± 1.0	87.3 ± 1.0	88.7 ± 1.1	2.0 ± 0.5	8.3 ± 0.5
Gray background	89.5 ± 0.4	88.1 ± 1.0	89.6 ± 0.6	1.4 ± 0.7	7.3 ± 1.1
Saliency penalty, $\lambda = 1$	88.7 ± 0.4	86.4 ± 0.3	87.8 ± 0.2	2.3 ± 0.5	7.7 ± 0.7
Saliency penalty, $\lambda = 10$	84.5 ± 1.6	81.5 ± 2.4	83.1 ± 2.2	3.0 ± 0.8	11.9 ± 2.1
Places365 swap (B, ours)	89.7 ± 0.2	89.1 ± 0.0	89.4 ± 0.4	0.5 ± 0.2	3.7 ± 0.2

Table 7: Six ways to train ResNet-18 with respect to the background, 3 seeds each.

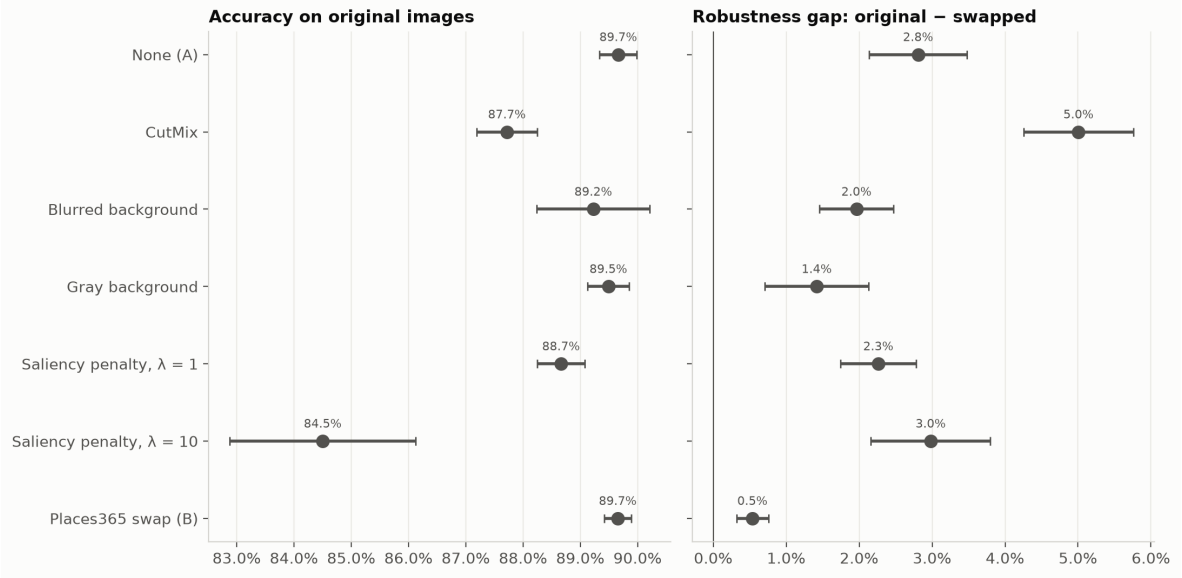


Figure 4: Original accuracy and robustness gap of the training strategies.

6.6 Where do the models look? Grad-CAM

All models put about 77–80% of their Grad-CAM mass on the pet, well above its 59% area, and A and B do not differ (Table 9). So Grad-CAM does not show the difference that accuracy, flip rate and probing show. This is a known limitation of Grad-CAM: it is coarse (7×7), it explains only the predicted class, and a model can look mostly at the pet while a small amount of background context still changes the decision on ambiguous images. Figure 6 shows such flips.

6.7 What do the features know about the background? Probing

ImageNet features encode the scene well (10–21%, i.e. 9–19× chance), which is expected: recognising scenes is useful for ImageNet. Fine-tuning on pets removes much of it (A: 3–6%), and background swapping removes most of the rest (B: 1.1–2.2%, close to chance; Table 10). Among the training strategies, every one that leaves a sizeable gap (gray, blur, CutMix, saliency penalty) also leaves 3.5–6.9% scene accuracy, while the random swap with $p \geq 0.25$ leaves 1.9–2.8%; CutMix, the strategy with the largest gap, also has the most scene information (6.9%). This is the mechanism behind H2: B learns features that are *invariant* to the background, so a new scene does not move them.

6.8 Transfer to real photos: Stanford Dogs

On real photos with natural backgrounds, B is at least as good as A for four of five architectures and significantly better for ResNet-50 (+0.7 p.p.), Swin-T (+1.0 p.p.) and the scratch model (+8.4 p.p.); for ResNet-18 the gain is +0.5 p.p. with a CI that touches 0 (Table 11, Fig. 8). ConvNeXt-T is the

Breed	Species	Gap A (p.p.)	Gap B (p.p.)
staffordshire bull terrier	dog	+28.1	+0.7
american pit bull terrier	dog	−28.0	−5.7
sphynx	cat	+11.3	+2.0
american bulldog	dog	+10.0	+0.0
abyssinian	cat	+7.5	+1.0
havanese	dog	+7.3	+1.7
wheaten terrier	dog	+7.3	+0.7
ragdoll	cat	+6.3	+2.0
chihuahua	dog	−5.7	−0.7
persian	cat	+5.7	+5.3
basset hound	dog	+4.7	+0.3
boxer	dog	+4.4	+0.0

Table 8: The 12 breeds with the largest |gap| for ResNet-18 A (average over seeds; 88–100 test images per breed, so one image ≈ 1 p.p.). Negative = more accurate on swapped images.

Model	CAM-on-pet, original	CAM-on-pet, swapped
ResNet-18 A ($p = 0$)	0.790	0.803
ResNet-18 B ($p = 0.5$)	0.798	0.803
ResNet-18, gray background	0.795	0.804
ResNet-18, blurred background	0.794	0.805
ResNet-18, CutMix	0.772	0.785
ResNet-18, saliency penalty $\lambda = 1$	0.783	0.795
ResNet-18, saliency penalty $\lambda = 10$	0.776	0.783

Table 9: Share of Grad-CAM mass on the pet (500 test images × 3 seeds). Pet area ≈ 0.59.

exception (−0.7 p.p., CI excludes 0), at the highest accuracy of all (95%). All alternative strategies are equal to or worse than A here, and the saliency penalty with $\lambda = 10$ loses 7.7 p.p. So the swap does not just teach the model to handle our composites; it does not hurt, and mostly helps, on a natural distribution shift. Cropping to the dog box lowers accuracy for all models (e.g. ResNet-18 A 83.8 → 69.6%). The likely reason is that many crops are small and are upscaled to 224 px, and that tight boxes cut off parts of the body and change the framing the networks were trained on, so this view is harder rather than "cleaner"; the A–B ordering on crops is the same as on full photos.

6.9 Case study: Waterbirds

Here the background is a real shortcut: it predicts the label 95% of the time in training. Plain fine-tuning reaches 89.0% on average but only 65.1 ± 10.5% on waterbirds photographed on land. The same swap remedy, unchanged (random scenes from our Places365 train pool, $p = 0.5$, automatic DeepLabV3 masks, no group labels), raises the worst-group accuracy to 80.5 ± 3.9% (+15 p.p.), the average accuracy to 94.1%, and the landbird-on-water group from 84.6 to 93.1%, while the majority groups stay at 93–99% (Table 12, Fig. 9). It also makes training far more stable: the worst-group std falls from 10.5 to 3.9 p.p.

6.10 Validation versus test

Validation accuracies (Table 14) are 2.5–4 p.p. higher than test accuracies for the pretrained models (e.g. ResNet-18 A 92.9% vs 89.7%) and 6–10 p.p. for the scratch models. Two reasons: the checkpoint was *selected* on validation (selection bias), and the validation set is small (552 images, ±1.1 p.p. binomial standard error). The ranking of models and the A/B gaps are the same on validation and test, which is the evidence that the test results are not a fluke of one split.

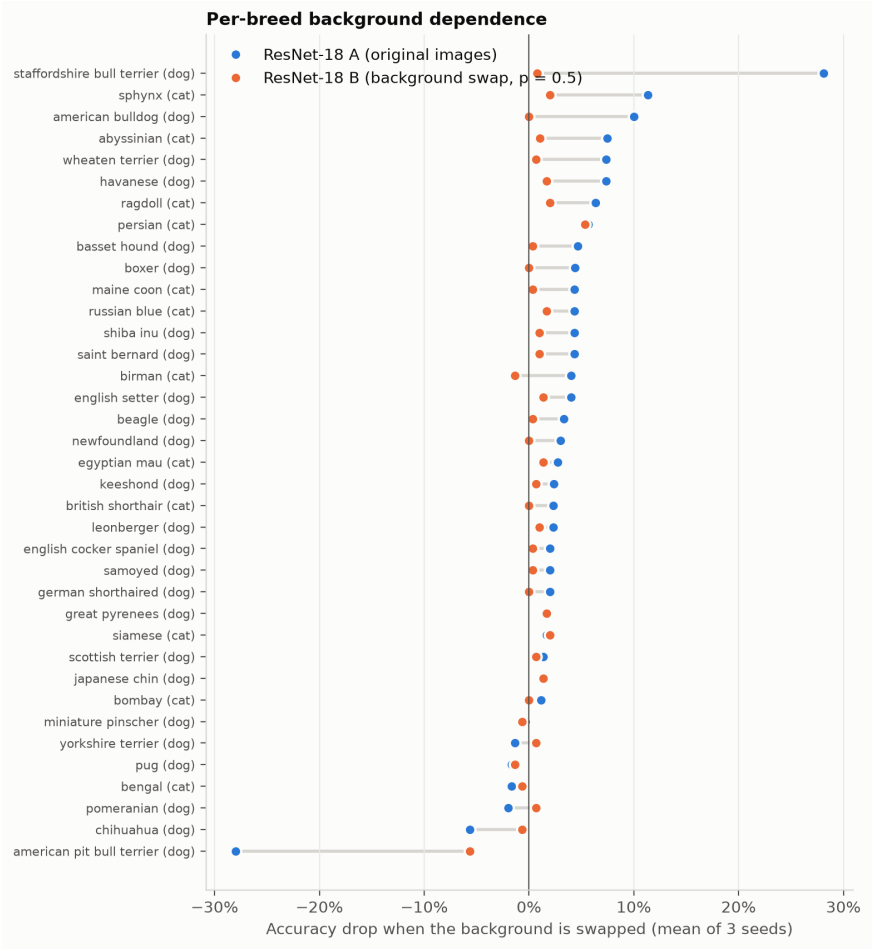


Figure 5: Per-breed gap of A and B.

7 Discussion

Distraction, not a breed–background shortcut (on Pet). Two observations together lead to this reading. (1) The bg-only accuracy of ResNet-18 is near chance, so the background alone says almost nothing about the breed to this model. (2) Yet swapping the background changes 8.4% of A's predictions. A model that ignored the background would change none; a model that used a breed–background association would also do well on bg-only. The consistent picture is that A's features mix pet and scene, and an unfamiliar scene shifts the features enough to push ambiguous images (Staffordshire vs pit bull) across a decision boundary. Probing supports this: A's features still encode the scene, B's mostly do not.

Why random scenes work and gray/blur do not fully. Gray and blur teach invariance to the *absence* of a background. The swapped test asks for invariance to the *presence of a wrong* background. Only random scenes provide training examples of that, and in thousands of variations (209 categories, a new draw every epoch). In probabilistic terms, the swap makes the background statistically independent of the label in the training distribution; the other strategies only weaken the dependence.

Why the saliency penalty fails here. RRR controls the first-order sensitivity $\partial p/\partial x$ around training images. A background swap is not a small perturbation: it replaces about 60% of the pixels. A model can be locally flat on the background and still respond strongly to a completely different scene. It also competes with the classification loss: at $\lambda = 10$ the model gives up 5 p.p. of accuracy and becomes *less* robust.

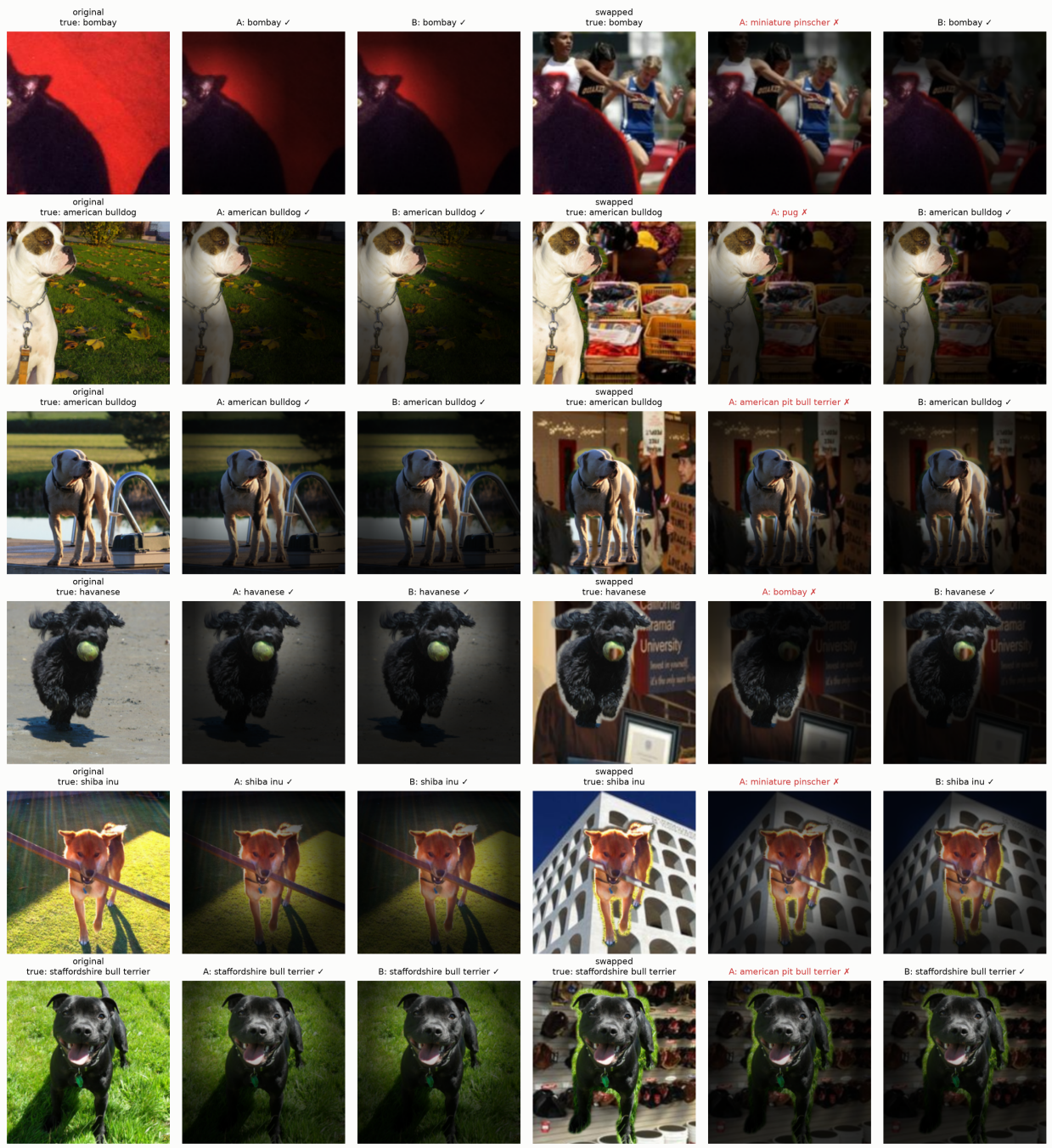


Figure 6: Images that every seed of A gets right on the original and wrong on the swapped version, while every seed of B gets both right. Columns: original image, Grad-CAM of A, Grad-CAM of B, swapped image, Grad-CAM of A (red = wrong prediction), Grad-CAM of B. Typical confusions: Staffordshire → pit bull, American bulldog → pug / pit bull.

Architecture	Before fine-tuning	A	B
ResNet-18 (ImageNet)	10.3	5.2 ± 0.2	2.0 ± 0.2
ResNet-18 scratch (random init.)	2.5	3.2 ± 0.3	1.9 ± 0.3
ResNet-50 (ImageNet)	15.9	6.4 ± 0.3	2.2 ± 0.2
ConvNeXt-T (ImageNet)	14.4	3.3 ± 0.3	1.1 ± 0.1
Swin-T (ImageNet)	21.5	5.7 ± 0.3	1.2 ± 0.2

(a) Architectures

ResNet-18 variant	Scene acc. (%)
ResNet-18 A ($p = 0$)	5.2 ± 0.2
ResNet-18, $p = 0.25$	2.8 ± 0.4
ResNet-18 B ($p = 0.5$)	2.0 ± 0.2
ResNet-18, $p = 1.0$	1.9 ± 0.1
ResNet-18, gray background	4.7 ± 0.2
ResNet-18, blurred background	4.8 ± 0.4
ResNet-18, CutMix	6.9 ± 0.3
ResNet-18, saliency penalty $\lambda = 1$	3.7 ± 0.5
ResNet-18, saliency penalty $\lambda = 10$	3.5 ± 0.2

(b) Training strategies

Table 10: Linear-probe accuracy (%) for the background scene category (87 classes, chance 1.1%), on penultimate features of swapped test images; 5-fold CV, mean ± std over seeds.

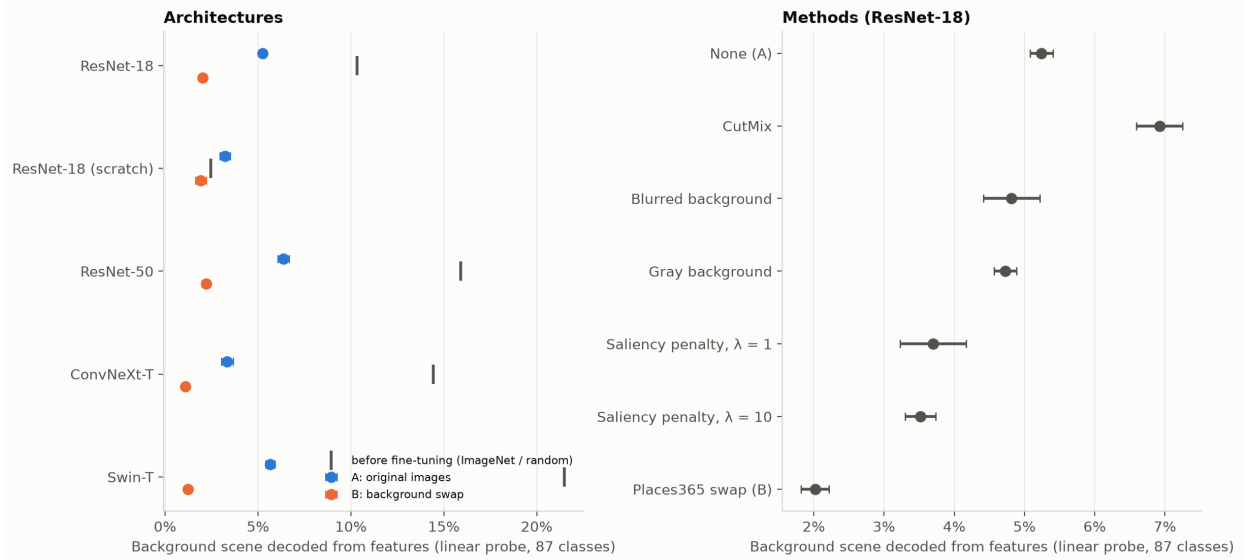


Figure 7: Background-scene probe accuracy per model.

Why bg-only is above chance for strong models. Bg-only images keep the shape of the pet as a gray silhouette. Ears, tail and body proportions are informative for breed (e.g. Basset hound vs Pomeranian). A strong model can recognise some breeds from the silhouette alone. B does not lower bg-only for ConvNeXt-T (9.6 → 9.2%), which is what we expect if the information is shape, since swapping preserves the shape. A silhouette-free control (e.g. inpainting the pet region) would separate shape from context; we leave it to future work.

Is there a price? On original images, no: B matches A within the seed spread for every pretrained architecture, and improves the scratch model. Costs are elsewhere: training needs foreground masks (available here, produced automatically by a segmentation network for Waterbirds) and about 35% more time per epoch for compositing (ResNet-18: 5.1 vs 6.9 min per run, Table 14).

ConvNeXt on Stanford Dogs. It is the only significant negative transfer result (−0.7 p.p.). Its A model is already the least background-sensitive on Pet (gap 1.0 p.p.) and the most accurate on Stanford Dogs (95.2%), and Stanford Dogs photos were part of its ImageNet pretraining data. One plausible explanation is that B moves ConvNeXt slightly further from its pretrained solution, which is especially good on exactly these photos. We report it as an honest exception.

Model	Full photo (%)	Dog crop (%)	Full, dogs only (%)	Pet test (%)	B−A full [95% CI] (p.p.)
ResNet-18 A ($p = 0$)	83.8 ± 0.6	69.6 ± 1.9	87.2 ± 0.3	89.7	
ResNet-18 B ($p = 0.5$)	84.3 ± 0.5	71.4 ± 0.8	87.8 ± 0.5	89.7	+0.5 [+0.0, +1.0]
ResNet-18 scratch A	40.7 ± 1.3	33.5 ± 1.5	47.5 ± 1.3	50.3	
ResNet-18 scratch B	49.0 ± 0.7	39.8 ± 0.3	55.4 ± 1.0	58.7	+8.4 [+7.3, +9.5]
ResNet-50 A	89.6 ± 0.6	74.1 ± 1.7	92.5 ± 0.6	91.7	
ResNet-50 B	90.3 ± 0.3	74.9 ± 2.0	93.1 ± 0.6	92.0	+0.7 [+0.3, +1.1]
ConvNeXt-T A	95.2 ± 0.4	91.0 ± 0.5	97.6 ± 0.3	94.4	
ConvNeXt-T B	94.5 ± 0.2	90.0 ± 0.2	97.2 ± 0.1	94.3	−0.7 [−0.9, −0.4]
Swin-T A	92.2 ± 0.4	85.2 ± 0.6	95.2 ± 0.3	93.5	
Swin-T B	93.2 ± 0.3	87.1 ± 2.3	95.7 ± 0.5	93.8	+1.0 [+0.6, +1.4]
ResNet-18, CutMix	82.2 ± 0.2	67.3 ± 2.4	85.5 ± 0.1	87.7	−1.5 [−2.1, −0.9]
ResNet-18, blurred background	83.6 ± 0.9	70.5 ± 0.2	87.2 ± 1.1	89.2	−0.1 [−0.6, +0.3]
ResNet-18, gray background	84.2 ± 0.2	70.6 ± 0.3	87.8 ± 0.5	89.5	+0.4 [−0.1, +0.8]
ResNet-18, saliency penalty $\lambda = 1$	82.3 ± 0.6	71.1 ± 0.4	86.8 ± 0.6	88.7	−1.5 [−2.0, −1.0]
ResNet-18, saliency penalty $\lambda = 10$	76.1 ± 1.4	65.7 ± 0.5	80.6 ± 1.0	84.5	−7.7 [−8.5, −6.9]

Table 11: External test: 3,730 Stanford Dogs photos of 21 Pet breeds, 37-way prediction ("dogs only" = argmax over the 21 breeds). The last column is a paired bootstrap over photos against the corresponding A model (for the strategies: against ResNet-18 A).

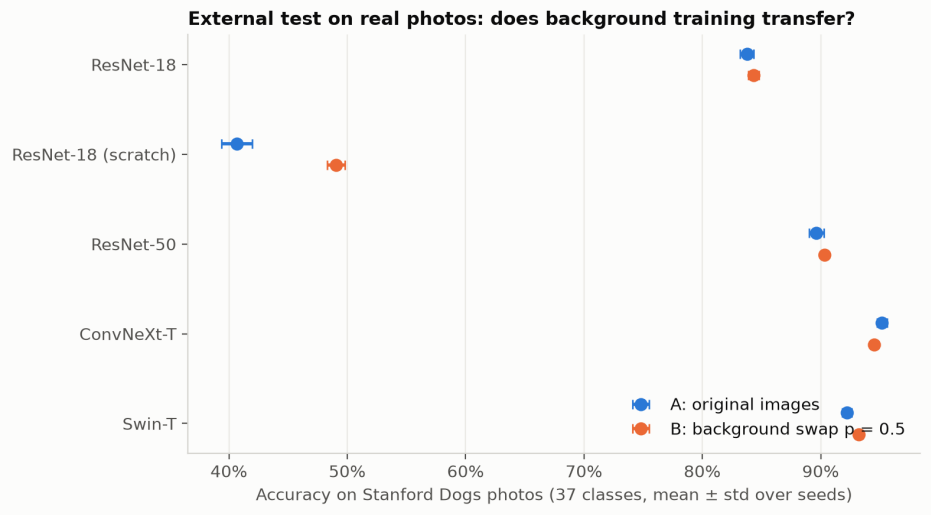


Figure 8: Stanford Dogs accuracy (full photos) of A and B for each architecture.

8 Limitations

- **Synthetic shift.** Swapped images are composites; despite soft masks they can show seams and lighting mismatch. We control for this with fg-only and check transfer on real photos, but a composite is not a real photo of a pet in that scene.
- **Masks.** Trimaps are human annotations with an uncertain border, which we assign to the pet; a few masks are almost empty. On Waterbirds, DeepLabV3 masks are imperfect (images with a mask below 0.5% of the area are left unswapped).
- **Silhouette leakage** in bg-only (see Discussion).
- **Hyperparameters** were tuned for ResNet-18 A and reused for all models; each architecture might do better with its own recipe. HOG's best C lies at the edge of the grid, so the baseline may be slightly under-tuned (it is not central to any claim).
- **Three seeds** give a rough estimate of training variance; std over 3 values is itself noisy. Bootstrap intervals cover test-set sampling, not training randomness.

Model (ResNet-18)	Average	Worst group	Landbird/land	Landbird/water	Waterbird/land	Waterbird/water
A: plain fine-tuning	89.0 ± 0.6	65.1 ± 10.5	99.3 ± 0.4	84.6 ± 4.6	65.1 ± 10.5	92.6 ± 2.4
B: random background swap	94.1 ± 0.1	80.5 ± 3.9	99.3 ± 0.3	93.1 ± 0.9	80.5 ± 3.9	93.1 ± 0.6

Table 12: Waterbirds test accuracy (%), mean ± std over 3 seeds. The worst group is waterbird on land for every run.

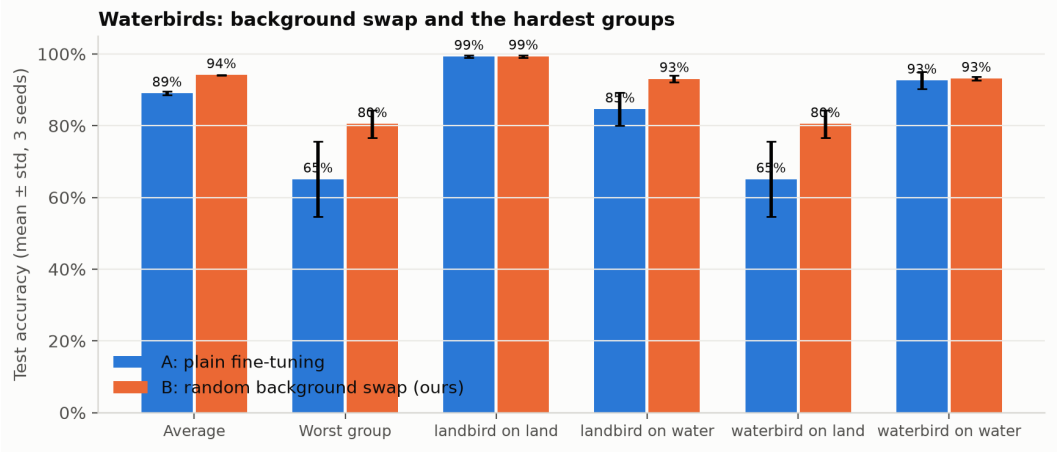


Figure 9: Waterbirds: average accuracy, worst-group accuracy and per-group accuracy of A and B.

- **ImageNet overlap** with Stanford Dogs inflates absolute external accuracies (but not the A–B comparison).
- **Scope.** One dataset family per question, one remedy family; no comparison with group-DRO or last-layer retraining on Waterbirds.

9 Conclusion

A standard fine-tuned ResNet-18 that classifies pet breeds with 89.7% accuracy changes 8.4% of its answers when only the background is replaced. It does not read the breed off the background; it is distracted by unfamiliar scenes, and for near-identical breeds that is enough to flip the decision. Replacing the background of training images by random scenes, already for a quarter of them, removes this sensitivity for every architecture we tried, costs no accuracy, strips background information out of the learned features, and works better than gray or blurred backgrounds, CutMix or a saliency penalty. The same simple fix transfers to real photographs and raises worst-group accuracy on the Waterbirds shortcut benchmark from 65% to 80%. The broader lesson for shortcut learning: test accuracy alone hides the problem; you have to break the correlation on purpose to see it, and if you can separate object from context (with masks or a segmentation model), randomising the context is a cheap and effective remedy.

Reproducibility and compute. Code, configs, splits, background manifests, per-image predictions of every run, training histories and the scripts that build every table and figure are in the repository. Every notebook runs end-to-end in Google Colab ("Run all"). The final set of 51 Pet runs and 6 Waterbirds runs was trained on a single local NVIDIA GPU; per-run times are in Table 14 (5–10 min for pretrained models, 14–18 min for scratch models). Checkpoints (≈13.5 GB) are stored outside git.

Team contributions. Amina Abdikalyk: project plan, splits, evaluation protocol, analysis, report. Islam Sanatbek: data pipeline, masks, background pools and compositing. Tomiris Rakymbay: HOG + SVM baseline, evaluation metrics, error analysis. Assylkhan Tleubayev: training

pipeline, model training, Grad-CAM.

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A Additional tables and figures

Model	Original	Swapped	Fg-only	Bg-only
HOG + linear SVM	9.1 ± 0.0	7.9 ± 0.0	5.2 ± 0.0	1.8 ± 0.0
ResNet-18 A ($p = 0$)	89.5 ± 0.3	86.8 ± 0.6	88.5 ± 0.8	2.1 ± 0.3
ResNet-18, $p = 0.25$	89.4 ± 0.6	88.9 ± 0.8	89.0 ± 0.7	1.8 ± 0.2
ResNet-18 B ($p = 0.5$)	89.5 ± 0.3	89.0 ± 0.0	89.3 ± 0.4	1.7 ± 0.2
ResNet-18, $p = 1.0$	88.6 ± 0.3	88.7 ± 0.4	88.6 ± 0.3	1.4 ± 0.3
ResNet-18, gray background	89.3 ± 0.4	88.1 ± 0.9	89.4 ± 0.6	2.2 ± 0.1
ResNet-18, blurred background	89.1 ± 0.9	87.3 ± 1.0	88.7 ± 1.0	2.2 ± 0.4
ResNet-18, CutMix	87.5 ± 0.6	83.0 ± 0.9	85.3 ± 0.8	2.6 ± 0.3
ResNet-18, saliency penalty $\lambda = 1$	88.6 ± 0.4	86.4 ± 0.4	87.7 ± 0.2	2.0 ± 0.3
ResNet-18, saliency penalty $\lambda = 10$	84.5 ± 1.6	81.7 ± 2.4	83.1 ± 2.2	1.8 ± 0.4
ResNet-18 scratch A	49.9 ± 1.6	42.5 ± 1.5	49.3 ± 1.2	2.6 ± 0.4
ResNet-18 scratch B	58.4 ± 0.9	57.8 ± 1.2	60.4 ± 1.4	2.0 ± 0.7
ResNet-50 A	91.6 ± 0.4	90.8 ± 0.2	91.2 ± 0.5	3.7 ± 0.4
ResNet-50 B	91.9 ± 0.6	91.6 ± 0.4	91.7 ± 0.3	3.3 ± 0.4
ConvNeXt-T A	94.3 ± 0.4	93.3 ± 0.5	93.9 ± 0.4	7.2 ± 0.7
ConvNeXt-T B	94.3 ± 0.2	93.9 ± 0.3	94.1 ± 0.2	7.1 ± 0.1
Swin-T A	93.4 ± 0.2	91.8 ± 0.4	93.1 ± 0.2	6.6 ± 0.8
Swin-T B	93.7 ± 0.2	93.4 ± 0.4	93.4 ± 0.3	4.7 ± 0.9

Table 13: Macro F1 (%) on the Pet test set, mean ± std over seeds.

Model	Epochs	Best epoch per seed	Val. original (%)	Val. swapped (%)	Final train acc. (%)	Min / run
ResNet-18 A ($p = 0$)	15	10, 7, 5	92.9 ± 0.2	89.3 ± 0.0	100.0	5.1
ResNet-18, $p = 0.25$	15	5, 7, 12	93.4 ± 0.4	91.5 ± 0.1	99.9	6.7
ResNet-18 B ($p = 0.5$)	15	7, 12, 10	93.5 ± 0.4	92.3 ± 0.1	99.8	6.9
ResNet-18, $p = 1.0$	15	7, 14, 7	92.8 ± 0.2	92.6 ± 0.6	99.8	9.7
ResNet-18, gray background	15	15, 5, 11	93.1 ± 0.5	90.9 ± 0.2	99.9	6.5
ResNet-18, blurred background	15	4, 9, 9	93.4 ± 0.5	90.3 ± 0.8	100.0	7.7
ResNet-18, CutMix	15	5, 13, 8	92.0 ± 0.9	84.7 ± 0.2	82.3	5.1
ResNet-18, saliency penalty $\lambda = 1$	15	8, 8, 5	92.2 ± 0.4	89.4 ± 0.6	100.0	10.5
ResNet-18, saliency penalty $\lambda = 10$	15	10, 13, 10	90.0 ± 0.5	85.3 ± 1.8	99.1	10.4
ResNet-18 scratch A	40	35, 40, 37	60.3 ± 1.3	47.3 ± 0.6	99.2	14.2
ResNet-18 scratch B	40	37, 40, 35	65.3 ± 0.8	64.4 ± 0.4	89.4	18.3
ResNet-50 A	15	8, 11, 5	95.6 ± 0.3	94.4 ± 0.6	100.0	6.6
ResNet-50 B	15	10, 13, 13	95.3 ± 0.5	94.9 ± 0.4	99.9	7.1
ConvNeXt-T A	15	3, 9, 12	97.0 ± 0.3	96.3 ± 0.3	100.0	6.6
ConvNeXt-T B	15	10, 6, 10	96.9 ± 0.4	96.8 ± 0.6	99.9	6.6
Swin-T A	15	8, 13, 9	95.7 ± 0.4	92.9 ± 0.2	99.8	7.6
Swin-T B	15	10, 15, 12	95.8 ± 0.5	95.4 ± 0.6	99.4	7.7

Table 14: Training summary. "Best epoch" is the selected checkpoint (1-based) for seeds 0, 1, 2; validation accuracies are those of the selected checkpoint. Train accuracy of CutMix is measured on mixed images against the hard label and is therefore lower.

Table 16: Every Pet run on the test set (%). Gap in p.p.

Model	Seed	Original	Swapped	Fg-only	Bg-only	Gap swap	Flip
HOG + linear SVM	–	9.8	8.5	7.3	3.8	1.3	74.3
ResNet-18 A ($p = 0$)	0	89.8	87.5	89.5	4.2	2.3	7.7
ResNet-18 A ($p = 0$)	1	89.9	86.3	88.0	3.3	3.6	9.4
ResNet-18 A ($p = 0$)	2	89.3	86.8	88.2	3.4	2.5	8.3
ResNet-18, $p = 0.25$	0	88.9	88.6	88.9	3.5	0.3	4.7
ResNet-18, $p = 0.25$	1	89.3	88.6	88.5	3.4	0.8	4.7
ResNet-18, $p = 0.25$	2	90.1	89.9	89.9	3.2	0.2	4.0
ResNet-18 B ($p = 0.5$)	0	89.5	89.1	89.0	3.0	0.3	3.7

Model	Seed	Original	Swapped	Fg-only	Bg-only	Gap swap	Flip
ResNet-18 B ($p = 0.5$)	1	89.6	89.1	89.6	2.9	0.5	4.0
ResNet-18 B ($p = 0.5$)	2	89.9	89.2	89.6	3.2	0.8	3.6
ResNet-18, $p = 1.0$	0	88.4	88.4	88.7	3.2	0.0	4.5
ResNet-18, $p = 1.0$	1	89.0	89.3	89.1	3.2	−0.3	3.9
ResNet-18, $p = 1.0$	2	88.8	88.8	88.5	3.2	0.0	4.8
ResNet-18, gray background	0	89.6	88.9	89.8	3.6	0.7	6.5
ResNet-18, gray background	1	89.1	86.9	88.9	3.7	2.1	8.6
ResNet-18, gray background	2	89.8	88.3	90.0	3.4	1.4	6.9
ResNet-18, blurred background	0	88.1	86.1	87.5	4.2	2.0	8.9
ResNet-18, blurred background	1	90.1	87.7	89.6	3.4	2.4	8.3
ResNet-18, blurred background	2	89.5	88.0	89.1	3.2	1.4	7.8
ResNet-18, CutMix	0	87.8	82.0	85.7	3.8	5.9	13.7
ResNet-18, CutMix	1	87.1	82.4	84.6	3.5	4.7	15.1
ResNet-18, CutMix	2	88.2	83.7	85.8	3.4	4.4	14.0
ResNet-18, saliency penalty $\lambda = 1$	0	89.1	86.5	88.0	4.5	2.6	8.0
ResNet-18, saliency penalty $\lambda = 1$	1	88.6	86.1	87.8	4.0	2.5	8.3
ResNet-18, saliency penalty $\lambda = 1$	2	88.3	86.6	87.6	3.8	1.7	6.9
ResNet-18, saliency penalty $\lambda = 10$	0	85.8	83.3	84.8	3.8	2.5	11.5
ResNet-18, saliency penalty $\lambda = 10$	1	82.7	78.8	80.6	3.3	3.9	14.2
ResNet-18, saliency penalty $\lambda = 10$	2	85.0	82.4	83.9	3.5	2.5	10.1
ResNet-18 scratch A	0	48.7	41.0	48.4	4.3	7.7	44.5
ResNet-18 scratch A	1	51.9	43.7	50.7	3.6	8.2	44.2
ResNet-18 scratch A	2	50.3	42.5	49.7	3.9	7.8	46.2
ResNet-18 scratch B	0	58.5	57.9	59.5	4.8	0.6	21.6
ResNet-18 scratch B	1	59.6	59.2	62.2	3.1	0.4	20.8
ResNet-18 scratch B	2	57.8	56.8	59.7	3.0	1.0	22.6
ResNet-50 A	0	92.1	91.0	91.8	4.9	1.0	4.6
ResNet-50 A	1	91.6	90.7	91.1	4.4	0.8	4.4
ResNet-50 A	2	91.5	90.9	90.9	4.9	0.5	4.6
ResNet-50 B	0	92.5	92.0	92.1	4.0	0.5	1.9
ResNet-50 B	1	91.5	91.4	91.5	4.3	0.1	2.2
ResNet-50 B	2	92.1	91.8	91.9	5.1	0.3	2.1
ConvNeXt-T A	0	93.9	92.8	93.5	10.3	1.1	3.3
ConvNeXt-T A	1	94.6	93.6	94.1	9.9	0.9	2.5
ConvNeXt-T A	2	94.6	93.6	94.4	8.7	1.0	3.5
ConvNeXt-T B	0	94.4	94.1	94.3	9.5	0.3	1.7
ConvNeXt-T B	1	94.1	93.7	94.0	9.1	0.4	1.7
ConvNeXt-T B	2	94.4	94.2	94.3	9.0	0.2	1.3
Swin-T A	0	93.3	91.5	93.0	7.1	1.8	5.5
Swin-T A	1	93.6	92.3	93.4	6.7	1.3	5.0
Swin-T A	2	93.6	91.9	93.1	8.4	1.7	5.9
Swin-T B	0	93.6	93.1	93.2	5.8	0.5	2.1
Swin-T B	1	94.1	93.9	93.5	6.6	0.2	2.0
Swin-T B	2	93.7	93.4	93.7	4.9	0.3	2.2

B Anticipated questions

Why not just evaluate on more test data instead of making composites?
Ordinary test data has the same background statistics as training data, so a background shortcut is invisible there. Composites let us change *only* the background and keep the pet pixel-identical, which turns a vague question ("does it use the background?") into a measurable difference between two inputs.

Are the swapped test backgrounds really unseen?
Yes, at two levels. The three background pools share no image, and they share no *scene category*:

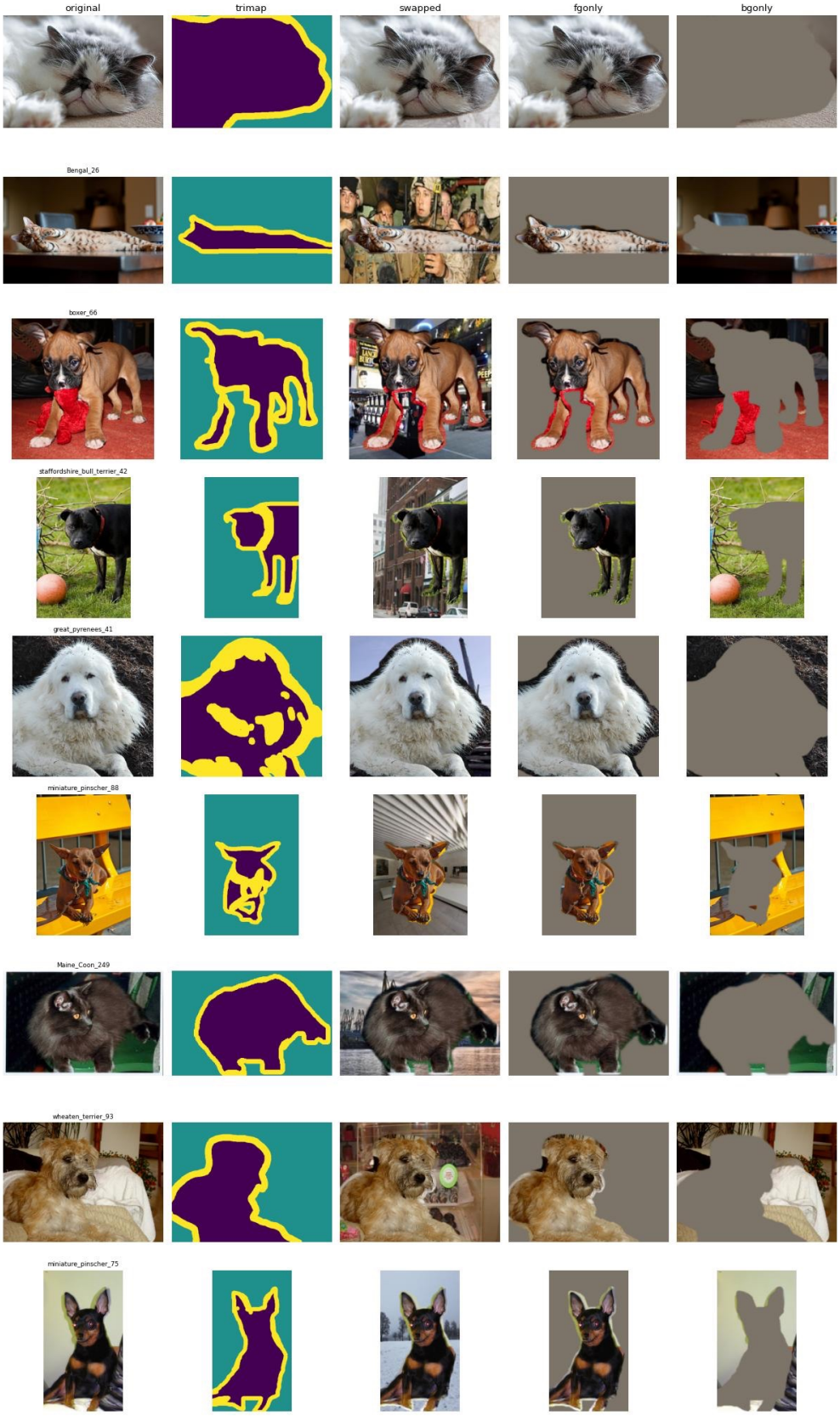


Figure 10: Examples from the data pipeline (first rows of the gallery made in week 1). Columns: original image, trimap (purple = pet, yellow = border, teal = background), pet on a new Places365 background (swapped), pet on the neutral gray background (fg-only), pet painted over with gray (bg-only).

Model	Seed	Average	Worst group	L/land	L/water	W/land	W/water
A	0	89.5	53.3	99.6	89.6	53.3	90.0
A	1	89.1	68.5	99.2	83.7	68.5	93.1
A	2	88.4	73.4	98.9	80.4	73.4	94.7
B	0	94.2	80.7	99.5	93.3	80.7	92.7
B	1	94.1	84.3	99.0	92.1	84.3	93.8
B	2	94.0	76.5	99.6	93.8	76.5	93.0

Table 15: Waterbirds test accuracy (%) per seed (L = landbird, W = waterbird).

the 87 test categories never appear in training (Table 2).

Could the gap come from paste artifacts and not from the background?
That is what fg-only controls. Fg-only images have the same seams and the same blending but a neutral background. A drops 1.1 p.p. there vs 2.8 p.p. on swapped images, so most of the drop is due to the scene. Also, B was trained with composites and might simply have learned to ignore seams; but B's fg-only gain over A (+0.8 p.p.) is smaller than its swapped gain (+2.2 p.p.), and B also helps on real Stanford Dogs photos, which have no seams.

Isn't B just seeing more data / more augmentation?
B sees the same images for the same number of steps; only the background of half of them is replaced. CutMix, gray and blur are also extra augmentations at the same rate, and none of them removes the gap. So it is the specific content of the augmentation, a random but realistic background, that matters.

Why choose the checkpoint on original validation accuracy?
To be fair to A. If we selected by swapped validation accuracy we would be optimising robustness on purpose, which would favour B. With selection on original images, both models are chosen for what the user would normally optimise.

Why $p = 0.5$ for B, if $p = 0.25$ already works?
$p = 0.5$ was fixed before the ablation as the natural midpoint. The ablation then showed that the result is not sensitive to p in $[0.25, 0.5]$, and that $p = 1$ starts to cost original accuracy.

Why is the flip rate larger than the gap?
The gap is a net change (wrong→right cancels right→wrong); the flip rate counts every change, including changes between two wrong classes. Flip $\geq$ |Gap| always.

How significant is the A vs B difference?
Across 3 seeds the gaps are 2.8 ± 0.7 vs 0.5 ± 0.2 p.p. Across test images the paired bootstrap gives 2.3 p.p., 95% CI [1.6, 2.9], with no bootstrap sample ≤ 0. For every architecture the CI excludes 0 (Table 6).

Why is bg-only accuracy near chance, but the gap still positive?
Because the model does not need the background to carry breed information to be disturbed by it. A new scene shifts the feature vector; for images near the boundary between two similar breeds, a small shift changes the decision. The per-breed results (Staffordshire vs pit bull) show exactly this.

Why does Grad-CAM not show a difference between A and B?
Grad-CAM has 7×7 resolution, explains one class, and measures where the gradient of the chosen class is large, not how much the decision would change if the background changed. Both models look mostly at the pet; the difference lies in how much the remaining 20% of attention can move the decision, which accuracy, flip rate and probing measure directly.

What exactly is the probe measuring and why on swapped images?
On swapped images the scene category is random and known for every image, so it is a clean label

that is independent of the breed. If a linear classifier can predict it from the penultimate features, the features contain background information that is useless for the task. Lower is better for our purpose.

Why does the saliency penalty fail?
It makes the model locally insensitive to infinitesimal changes of background pixels around training images, but a background swap changes 60% of the pixels completely. Local flatness does not imply invariance to large changes, and at $\lambda = 10$ the penalty fights the classification loss (accuracy -5.2 p.p.).

Why does CutMix make things worse?
CutMix is not mask-aware: it pastes patches over the pet as well and mixes labels by area. The model learns that a part of the image can belong to another class, which makes it more sensitive to local content anywhere, including a new background. It is a regulariser for accuracy, not for background invariance.

Why do stronger models have smaller gaps?
Better ImageNet features describe the object more completely, so the pet evidence dominates. This matches Xiao et al. [20]. But the gap does not vanish (Swin-T A still flips 5.5%), and B still reduces it 3–5×.

Why train a model from scratch at all?
To see the effect without ImageNet knowledge. With only 3,128 images the scratch model relies on the background most (gap 7.9 p.p., flip 45%), and swapping both removes the gap and raises accuracy by 8.4 p.p.: for small datasets the remedy is also a strong regulariser.

Is Stanford Dogs a fair external test, given ImageNet pretraining?
For absolute accuracy, no: pretrained backbones saw these photos. For the comparison A vs B, yes: both start from the same weights. The scratch models never saw them, and there B gains 8.4 p.p.

Why is accuracy on dog crops lower than on full photos?
The crops are often small and low-resolution after cropping and upscaling to 224 px, and they lose context such as the dog's full body. Our models were trained on photos in which the pet usually fills a large part of the frame but not all of it. So the crop view is harder, not "cleaner"; it is reported for completeness.

How were Waterbirds masks obtained without annotation?
A pretrained DeepLabV3-ResNet50 segments the "bird" class (VOC label set). The mask is soft-blurred like Pet masks; if it covers less than 0.5% of the image, the image is kept original. No group labels are used anywhere in training.

Why is the Waterbirds worst group waterbird on land?
In training, 95% of waterbirds are on water. A model that uses "water" as evidence for "waterbird" then sees land and votes "landbird". Waterbirds are also the minority class (about 23% of training images), so this group is both rare and contradicted by the shortcut.

What is the difference between this and group DRO?
Group DRO [17] needs group labels (bird type *and* place) and reweights the loss towards the worst group. Our remedy needs only a foreground mask (automatic here) and changes the data. We did not run group DRO; on standard ResNet-50 Waterbirds setups it reaches higher worst-group accuracy, so our result shows that a simple, label-free data fix gets a large part of the way.

Why are validation accuracies higher than test accuracies?
Checkpoints were selected on validation (selection bias), and validation has only 552 images. The ranking and the A/B gaps agree between validation and test.

Why only 3 seeds?
Compute budget: 51 Pet runs plus 6 Waterbirds runs. Three seeds are enough to see whether an effect is larger than training noise, and the bootstrap adds a test-set uncertainty estimate.

What would you do next?
A silhouette-free bg-only control (inpainting), Mixed-Same composites (backgrounds from the same breed) to test a breed–background association directly, a per-architecture hyperparameter search, and group-DRO and last-layer retraining baselines on Waterbirds.

C Glossary

Shortcut / spurious feature A feature correlated with the label in the training data that the model uses instead of the intended one.

Trimap Per-pixel annotation: pet / background / uncertain border.

Alpha blending $\alpha x + (1 - \alpha)b$: mixing two images with a per-pixel weight.

Fine-tuning Starting from pretrained weights and training all layers further on the new task.

Backbone / head The feature extractor / the final linear classifier.

Epoch / step One pass over the training set / one optimizer update on one batch.

Mixed precision (AMP) Computing in 16-bit floats where safe, with loss scaling to avoid underflow.

Gap Accuracy on original minus accuracy under a condition, in percentage points (p.p.).

Flip rate Share of images whose prediction changes between two conditions.

Percentage point (p.p.) Difference of two percentages: $89.7\% - 86.9\% = 2.8$ p.p.

Bootstrap CI Uncertainty estimate obtained by resampling the test images with replacement.

Grad-CAM Heat map of the image regions whose features raise the predicted class score.

Linear probe A linear classifier trained on frozen features to measure what information they contain.

Worst-group accuracy The lowest accuracy over predefined subgroups of the test set.

Distribution shift A test distribution that differs from the training distribution.